

VR-Forms. Preprint.

Part IV. Transits

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IV.0. Introduction

Part III established the conservativity of VR-Forms over VR-Sets. This is a negative result: the formal register **does not generate** new ontological statements. Part IV provides the positive content: the formal register **helps** in reasoning about operational sets, even without adding new theorems.

The help consists in the following. Many classical proofs use uncountable intermediaries: classical $\mathbb{R}$, σ -algebras, Lebesgue measure, the axiom of choice for uncountable families. These intermediaries are not themselves operational. But if a proof using them ends in an ontological statement, then — by conservativity — this statement is provable also without the intermediaries. Part IV formulates the **transit**: a rule by which a formal proof of an ontological statement is recognised as a proof in VR-Sets.

Contents: notion of transit (§IV.1); transit rule and its justification (§IV.2); notion of witness (§IV.3); safe and problematic transits (§IV.4); examples (§IV.5); summary (§IV.6).

IV.1. The notion of transit

A transit is the passage from the formal register to the ontological. More precisely: the use of formal terms as intermediate means in a proof whose result is an ontological statement.

Definition IV.1 (transit)

A **transit** is a derivation in T_1 in which:

- (a) the final formula φ is ontological ($\varphi \in L_0$);
- (b) in intermediate steps of the derivation formal terms are used (a nonzero number of steps contains the marker $\ulcorner \cdot \urcorner$).

A transit, then, is a derivation that **passes through** the formal register on the way to an ontological result.

What conservativity promises

By Theorem III.1, if such a derivation exists in T_1 , there also exists a derivation of φ in T_0 (without formal terms). That is, every transit **can be rewritten** as an ontological proof.

However, the rewriting may require substantial work. In practice, a formal derivation can be shorter and clearer than the equivalent ontological one. A transit is a way of **using** the formal register to ease the work, knowing in advance that the result is ontological.

IV.2. The transit rule

Let us formalise how exactly to use a transit.

Transit rule

If a derivation D ending in an ontological formula φ is constructed in T_1 , then φ is accepted as proved in T_0 .

Justification. By Theorem III.1. The existence of a derivation in T_1 of an ontological formula is equivalent to the existence of a derivation in T_0 .

The transit rule, then, is not a new inference rule. It is a **methodological permission**: one may work in the formal register, and the result automatically belongs to the ontological, provided it is itself ontological.

What transit does not do

A transit does not turn formal terms into ontological objects. The Russell term $\ulcorner R \urcorner$, used in a transit, does not become a set. It remains a formal term used as an intermediate syntactic element. Only the final formula, if it is ontological, passes into the ontological register.

A transit does not allow anything to be proved in T_0 that could not be proved without the transit. It is only a **convenient way** to conduct a proof. No new strength is provided by the transit.

A transit does not «justify» formal terms. That a certain formal term is used in a transit does not mean it is operationally realisable (see §II.7). Many formal terms figure in transits while remaining non-realisable. Realisability is a separate question; transit is a syntactic procedure.

IV.3. Witness

Connected with transit is the notion of **witness** — an operational object whose existence is proved by the transit.

Definition IV.2 (witness)

Let a formula φ have the form $\exists x \psi(x)$, where ψ is ontological. A **witness** of φ is an operational set A such that $\psi(A)$ is true in T_0 .

A transit may prove $\exists x \psi(x)$ — that is, the existence of a witness — without exhibiting a specific A . This is the typical situation of non-constructive proofs in classical mathematics.

Extraction of a witness

If a transit yields $\exists x \psi(x)$, then by conservativity there is a derivation of $\exists x \psi(x)$ in T_0 . However, this derivation may not be constructive: it establishes existence without exhibiting x .

The existence of a **witness extraction** is a separate question. For constructive proofs existence entails construction; for classical it does not necessarily. VR-Forms inherits this

distinction: transit yields an ontological statement but does not always yield an operational construction.

Remark

This is the place where the boundary between two possible roles of VR-Forms is visible.

Weak role (proof of existence): transit is used to establish that an operational object with a given property exists. The object itself may remain unspecified.

Strong role (construction): transit is used as a heuristic — formal reasoning suggests how to build an operational object. The construction is then verified in the ontological register.

Both roles are admissible. In practice the strong role is preferable, since it yields an explicit operational object, not leaving it «known only formally».

IV.4. Safe and problematic transits

All transits are safe in the sense of consistency (Part III). However, from a practical viewpoint **direct** and **problematic** transits can be distinguished.

Direct transits

A transit is called **direct** if in it formal terms are used as convenient abbreviations for ontological objects. That is, all intermediate formal terms are operationally realisable (§II.7), and the use of the formal register is mere syntactic convenience.

Direct transits trivially rewrite into ontological proofs. Their use is a stylistic decision.

Problematic transits

A transit is called **problematic** if it essentially uses **non-realisable** formal terms. That is, intermediate steps refer to $\mathbb{R}$, $\wp(\mathbb{R})$, «the Vitali set», to choice functions over uncountable families, and so on.

Here one must be more careful. Conservativity guarantees that the result of a problematic transit is translatable into the ontological register — but the rewriting may be cumbersome, and sometimes require substantial operational work.

A principal observation

Problematic transits are places where classical mathematics essentially leans on uncountability. If the result of a transit is ontological (for example, a statement about a concrete computable function), then by conservativity it is provable also without uncountable intermediaries. **This is the programme of VR-Audit** — analysis of classical theorems with the aim of extracting the ontological remainder.

Within Part IV: established only that the rewriting is possible. **How** to conduct it in concrete cases — the subject of Parts V onward, and in full — the programme of VR-Audit.

IV.5. Examples

Example 1: proof of the countability of the set of algebraic numbers

Classical proof: the set of polynomials with rational coefficients is countable (as a countable union of countable sets); each polynomial has finitely many roots; therefore there are countably many algebraic numbers.

This proof is already ontological: all its steps concern operational sets ($\mathbb{Q}$, polynomials with rational coefficients, roots). No transit is needed. The result belongs to T_0 .

Example 2: proof of Bessel's inequality

Bessel's inequality in Hilbert space: for any orthonormal family $\{e_n\}$ and any x in the space, $\sum | \langle x, e_n \rangle |^2 \leq \|x\|^2$.

The classical proof uses the formulation «Hilbert space», which in the full classical form is uncountable. However, the inequality itself is a statement about operational objects, if x and $\{e_n\}$ are computable. Transit: using $\ulcorner$ Hilbert space $\urcorner$ as a formal term in intermediate steps, we obtain an ontological result — the inequality for concrete computable x and $\{e_n\}$.

By conservativity this means that Bessel's inequality for computable data is provable in VR-Sets directly. The transit here is convenience, not necessity.

Example 3: a proof essentially using the uncountable

The Hahn–Banach theorem in its general classical formulation: every bounded linear functional given on a subspace extends to the whole space with preservation of norm.

The classical proof uses Zorn's lemma, which depends on AC for uncountable families. The formal term $\ulcorner$ functional on the whole $\mathbb{R}$ -space $\urcorner$ is essentially non-realisable.

Transit: if the result of the theorem is a statement about a computable extension of a computable functional, then by conservativity the result is provable in VR-Sets. But the rewriting here is non-trivial: an explicit operational construction of the extension is needed, which in the general case requires additional work. This is an example of a **problematic transit**, requiring operational labour to extract the ontological content.

Example 4: spectral theorem for the harmonic oscillator

The harmonic oscillator has discrete spectrum $E_n = \omega(n + 1/2)$. The classical formulation of the spectral theorem uses $\ulcorner$ projection-valued measure $\urcorner$ — a formal term requiring an uncountable σ -algebra.

But for the harmonic oscillator this formal apparatus is superfluous: the discrete spectrum and eigenfunctions (Hermite polynomials multiplied by a Gaussian) are operational. The transit here yields an ontological statement — for example, computability of matrix elements of the energy operator — without essential dependence on uncountable structure.

This is an example where the formal register is **replaced** by a direct operational construction. The transit here is a methodological permission to use the classical proof, but more valuable content is provided by the direct operational derivation, which we build directly in T_0 .

IV.6. Summary of Part IV

Established:

- (1) Transit as a derivation in T_1 passing through formal terms and ending in an ontological formula (Definition IV.1, §IV.1).
- (2) Transit rule: every transit yields a proof in T_0 (§IV.2). Justification — Theorem III.1.
- (3) The notion of witness and the distinction of the weak (existence) and the strong (construction) roles of transit (§IV.3).
- (4) Distinction of direct and problematic transits (§IV.4); problematic — places where classical mathematics essentially leans on the uncountable.
- (5) Examples: algebraic numbers (no transit), Bessel's inequality (direct transit), Hahn–Banach (problematic), harmonic oscillator (direct operational replacement) (§IV.5).

Methodological role of Part IV. Part IV closes the technical apparatus of VR-Forms. Onward we pass to substantive **application** of the apparatus to classical mathematical (Part V) and non-mathematical (Part VI) formal terms.

Connection with VR-Audit. Part IV outlines a programme: for each classical theorem φ , determine whether its proof is a direct transit (i.e. φ is provable in VR-Sets without essential loss), a problematic transit (requires operational labour to extract content), or not a transit at all (φ is non-ontological, remains in the formal register). Systematic conduct of such an analysis is the content of the future work VR-Audit; VR-Forms provides for it the apparatus.